



Problem # 1: Multiple choice question: [12 x 1pt]

- Which of the following instructions is the same as **JB** instruction?
A. JNGE B. JNG C. JNAE D. JNA
- Find the content of **AL** and **ZF** register after execution of the following code:
MOV AL, FCH
TEST AL, 16
A. AL=FCH; ZF=1 B. AL=FCH; ZF=0 C. AL=00H; ZF=1 D. AL=0CH; ZF=0
- Instruction pointer (IP) contains the offset address of _____ segment.
A. Data segment B. Code segment C. Stack segment D. Extra segment
- The output of the following operation on register **BL** is:
MOV BL, FCH
MOV CL, 8
ROL BL, CL
A. BL=CFH B. BL=FCH C. BL=0FH D. BL=00
- If the range of physical addresses of a given memory is **0000h-3FFFh**, the number of the required address bus Bits (pins) in this memory is:
A. 16 bits B. 14 bits C. 12 bits D. 15 bits
- In the following instruction sequence, the values of the **parity**, **auxiliary carry**, and **sign** flags are:
MOV AX, 0C17Fh
ADD AX, 3
A. PF=0, AF=1, SF=0 B. PF=1, AF=1, SF=1 C. PF=0, AF=0, SF=1 D. PF=1, AF=0, SF=0
- If the offset address of **X** in data segment is equal to **323H**, and **DS=2500H**, then the physical address of **X** is:
A. 20215 H B. 25323H C. 2500 H D. 323000
- Which of the following is an example of a based indexed addressing mode?
A. MOV AX, [BP + SI + 1Ch] B. MOV AX, [BP+1] C. MOV AX, [SI+1] D. MOV AX, [BX]
- Suppose **X** defined as below; what is the content of **CX**, and **AH** registers after execution of the following instructions:
X DW 4580H, 0ABCH, 1100, 2222H
MOV BX, OFFSET X
INC BX
MOV CX, [BX]
INC BX
ADD AH, [BX]
A. CX=BC45H; AH=BCH B. CX=0ABCH; AH=0AH
C. CX=BC00H; AH=B0H D. CX=BC45H; AH=0CH
- Assuming **DS=4000h**, **SI=10h**, and **BX=23h**, the physical address of the source operand in the instruction **MOV AX, [BX + SI + 0C3h]** is:
A. 400F6h B. 4FFF6h C. 40033h D. 40123h
- The value of **AX** at the end of execution of the following 8086 assembly instruction sequence is:
MOV AX, -9
SAR AX, 8
A. AX=00FFh B. AX=0FFFh C. AX=FF00h D. AX=0FFF0h
- If **AL** contains the ASCII code of an uppercase letter, it can be converted to lowercase by:
A. XOR AL, 32 B. AND AL, 00100000b C. OR AL, 00100000b D. SUB AL, 32

*****Write the answers of Problem # 1 in the table below:

Q#1	Q#2	Q#3	Q#4	Q#5	Q#6	Q#7	Q#8	Q#9	Q#10	Q#11	Q#12	
A	A	B	B	A	B	B	A	A	C	B		12

Problem # 2: Find the contents of the registers written below for each of the following assembly codes. [10 pts]

A) X1 DW 3000h
X2 DW -2, 27ABh, 0CCCCCh
X3 DB 116
MOV AL, X3
PUSH X1+1
PUSH X1+4
PUSH X1+2
POP DX
POP CX
POP BX
HLT

AL=
CX=

BX=
DX=

X1	00
X1+1	30
X2	FD
	FF
	AB
	27
	CC
	CC
X3	0B

B) X DB 1,2,3,4,5,6,7,9
MOV CX, 8
MOV SI, 0
L2: CMP X[SI], 6
JA L
ADD AL, X[SI]
L: INC SI
LOOP L2

AL=

SI=

C) MOV BL, 13H
L2: SHR BL, 1
JNC L
INC DL
L: CMP BL, 0
JNZ L2
XOR DL, FFH

BL=

DL=

D) MOV CX, 8
MOV BL, 2
MOV BH, 0
MOV AL, 5
L: MOV AH, 0
DIV BL
CMP AL, 0
LOOPNZ L

AX= 01 02

BX= 00 02

CX= 00 08

Problem # 3:

- A) Write an 8086 assembly program to find the total number of the memory byte locations in the memory array named Array in which the bit 2 is set then store it in the byte memory location named count. [4 pts]
Array DB 21H, 84h, 0F7h, 0E8h, 0ABh, 8Ch, 0C4h, 44h, 74h, 95h
count DB ?



B) Implement the following if statement logic using assembler codes: [2 points]
If $x < 20$ $s = a + 10$ else $s = b + 20$; assuming x, s, a, b signed memory locations.

```
org 1000h  
cmp 8, 10  
JGEL2  
ADD S, 20
```

```
hlt  
$DB0  
x DB 1
```